

Haptic Scores

Non-Visual Ways of Experiencing Choreography

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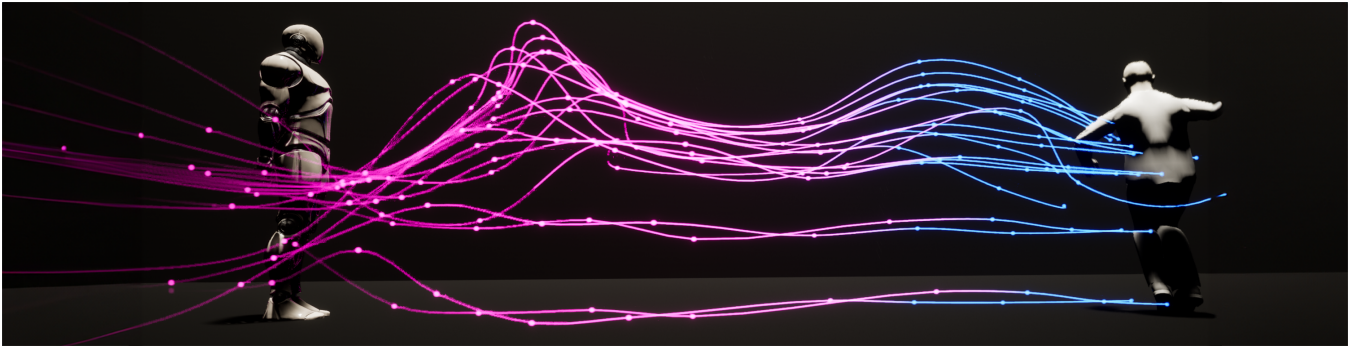


Figure 1: UE5 screenshot displaying the variation in the position of the particles while the source body is in motion. As the avatar moves (source body, right), the particles emitted at regular intervals create spatial configurations that result in unique rhythmic patterns colliding with the receptive body (left).

Abstract

Current developments in haptic interfaces predominantly aim to achieve more ‘realistic’ sensations with a particular emphasis on the sense of touch, focusing on the fingertips for manipulating digital objects in VR/AR experiences. However, these haptic sensations are usually complementary to the visual stimuli due to HCI biases toward the sense of sight. This research aims to develop an approach for the development of a standalone haptic experience by transforming pre-recorded human movement into haptic feedback. Using artistic and design practice research with *Teslasuit* as a principal interface, this research investigates how mocap data and electrical stimulation can be used to develop haptic scores by redirecting movement traces from a moving body to another body. This approach to movement transformation presents a way in which human movement, particularly dance, can be experienced from an audience perspective beyond sight, expanding existing research on using electric stimulation for creative practices.

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CCS Concepts

• **Human-centered computing** → **Interaction design process and methods; Empirical studies in HCI.**

Keywords

Dance, EMS, TENS, Haptic Experiences, Motion Capture, Soma Design

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1 Problem Statement

There is a growing interest in the research community towards the use of haptic interfaces in applications outside of medical and scientific fields. Current developments predominantly aim to achieve more immersive or ‘realistic’ sensations with a particular emphasis on the sense of touch, specifically focusing on the fingertips for manipulating digital objects in VR/AR experiences. However, these haptic sensations are usually complementary to the visual stimuli due to HCI biases toward the sense of sight. Furthermore, focusing only on reproducing touch-like sensations that mimic able-bodies’ perception qualities hides the potential that these devices hold for developing innovative approaches to interaction design. This work

aims to develop an approach for the development of a standalone haptic experience. In transforming pre-recorded human movement into haptic feedback, we will emphasise that touch can be experienced and explored without the addition of visual stimuli. Centred on the dynamic relationship between a moving body and the receiver of mediated movement traces, this work seeks to address critical questions: (1) What theoretical and practical approaches can be taken in the transformation of mocap data into haptic experiences? (2) What significance do haptic devices and electrical stimulation hold for artistic practices? (3) How do these practices extend into the development of non-visual approaches to aesthetics and interactions?

These inquiries guide the development of Haptic Scores, an approach that is part of a series of explorations focused on designing experiences for haptic perception, specifically utilising electric stimulation in artistic practices.

2 Related Work

The interest in the development of haptic interfaces and devices increased in the last decades, particularly in HCI, social sciences, design [13], and affective system design [29]. One of the principal functions of these interfaces is to improve the communication between humans and machines by focusing on the sensory-motor human skill [10]. Adilkhanov et al. identified four main application domains of haptic devices developed during the period 2010-2021: teleoperation, training/education, guidance/notifications, and entertainment [1]. In the ninety devices reviewed by Adilkhanov et al., no mention is made of using haptic devices in artistic or design contexts, evidencing a research gap in the uses, possibilities, and influences that haptic feedback devices could have in multimodal artistic practices.

Numerous haptic technologies have been developed to create immersive and realistic interactions [1] [20] [28]. Commonly employed haptic interfaces involve force feedback and vibrations. Another approach to haptics is the use of electrical muscle stimulation (EMS) and transcutaneous electrical nerve stimulation (TENS). Based on electrical impulses that are sent into the body, these impulses produce different sensations by manipulating four variables: pulse width, frequency, duration, and current intensity. While primarily used for muscle and nerve rehabilitation, this technology has increased its presence in the field of human-computer interaction (HCI) due to the possibility of generating sensations without the use of bulky mechanical actuators [17]. It also offers the possibility to develop innovative sensations and applications outside of the medical field, such as temperature sensations in head-mounted displays (HMD) [21] or improving dexterity for playing instruments [25]. The following section will review the historical use of EMS and TENS, focusing on their application in the arts. This review will position this research as an essential contribution to the exploration of EMS and TENS, using state-of-the-art technologies involving interdisciplinary methods drawn from art and design.

2.1 Electricity and the Human Body in Performing Arts

In 1747, Jean Jallabert discovered that electricity can trigger independent muscle contractions. His discovery led to the recovery of

movement in a patient with a paralysed arm [8]. Since then, the performative potential of Jallabert's discovery in causing involuntary muscle movement has remained a topic of interest in contemporary performative art practices. Three aspects emerge as a core of interest between the multilayered facets for analysis and exploration: the body being manipulated by an external force and, with this, questioning agency upon our physicality; the role of technology and the possibilities to intervene in our own or someone else's body; and the aesthetic outcomes of such interventions.

2.2 Contemporary Electric Performance Art

Numerous artists and performers produced works aligned with the early works of Duchenne de Boulogne in 1854 using single muscle contractions by the means of electrodes placed on the body. Among those artists, we find: Stelarc [23], Arthur Elsenaar [7], Choy Ka Fai [4], Isaiah Wilson [31], Stahl Stenslie [24], and Pedro Lopes [18]. These artists' works, except for Stenslie's, focus on manipulating a body with an EMS or TENS interface. Similar to the actions carried by a puppeteer, a body moves as a result of someone else's decisions (human or machine) transferred from an input device, causing muscle responses in a body that is expecting to receive the stimulation. The perception of the viewer is challenged by the apparent displacement of the human agency upon the movement of its own physicality to an external input source. The case of Stenslie's work is different; there is no performer or manipulated body to observe, and the haptic experience is shared between participants or audience members. The focus of Stenslie's *Inter_skin* (1994) is the exchange of mediated touch; as one participant touches themselves, another participant receives the touch sensation through the device onto their body. The haptic experience is placed on the participant/audience's bodies for them to perceive instead of observing a body being manipulated. This research is aligned with this approach to the use of EMS and TENS in artistic contexts. The appreciation of the experience is then a non-visual dependent instance where the relation between bodies, interfaces, and sensations defines the media of artistic appreciation and perception.

3 Methodology

This work will follow the explorative sets of methods from soma design as defined by Höök, which include: "(1) ideation methods, (2) engagements with the experiential and aesthetic potential of the digital and technological materials, and (3) evaluation approaches for soma design once they start to take shape" [12]. The combination of methods during those three stages will serve as a base for creating a set of explorations that will help define, test, and evaluate the designed interactions.

For those explorations, this research will use *Teslasuit* as a principal device that allows for data collection (mocap) and provide haptic sensations to the wearer. This combination of technologies (mocap and haptic feedback) makes *Teslasuit* the appropriate interface for facilitating, through practical work, access to valuable information from the moving body that will help to explore possible answers to the research questions.

3.1 The Interface

The Teslasuit, developed by VR Electronics, is a two-piece wearable bodysuit consisting of trousers and a jacket [26]. The suit is equipped with 14 Inertial Measurement Units (IMU) that monitor and capture the wearer's movements. Employing EMS and TENS, the Teslasuit delivers haptic feedback via eighty embedded electrodes. This technology enables electrical signals of varied intensity and quality to be applied, allowing the wearer of the suit to perceive realistic sensations that enhance immersive experiences.

The range of feedback sensations and potential applications of the Teslasuit are currently in the early stages of development. Existing research has explored the medical domain, particularly focusing on rehabilitation and training [3] [9] [11] [30]. Additionally, the scope of research has also extended to the realms of the gaming industry and augmented/virtual reality (AR/VR) [16] and cinematic experiences [2]. However, it is essential to note that within the existing literature, the Teslasuit is predominantly acknowledged as a promising area for further exploration [15] [19] [27], leaving the possible applications of Teslasuit in performing arts and design yet to be further researched.

3.2 The Sensation

The haptic feedback comprises three variables from which all the different haptic sensations (in game engines, also named haptic materials) for the Teslasuit can be created: (1) Pulse width: This variable correlates with the intensity of the current. A higher pulse width results in a greater intensity of EMS, leading to muscle engagement. (2) Frequency: This variable relates to the smoothness of the sensation. A higher frequency, indicating more waves per millisecond, gives the sensation of a continuous current. In contrast, lower frequencies translate into granular or rough sensations. (3) Duration: This variable determines how long the haptic feedback is applied. Short duration could evoke a tap or hit sensation, while longer duration may create the feeling of walking through water, for example. The appropriate combination of pulse width and frequency can even induce involuntary movements, where sufficient electricity flow causes muscles to contract, resulting in corresponding body movements. Through the combination of these variables, we can design different haptic materials that aim to resemble known sensations. However, exploring these parameters allows for the creation of sensations that are particular to the interface without resembling any previously known or expected haptic experience.

4 Haptic Scores

4.1 Inspiration

In the first instance, a short dance sequence was captured using Teslasuit. This dance sequence is used as a source for the transformation of movement data into haptic feedback. The created choreography has a variety of movement qualities that help to test the influence that they have in the development of haptic experiences (slow and fast movements, on the place and around the space, different levels like standing or kneeling on the floor, and small and big gestures).

The Teslasuit's Studio software facilitated the retrieval of mocap data in three different formats: raw data (.csv file) and processed data (.csv and .fbx files). The recorded movement was visualised by applying the processed data (.fbx file) to a digital avatar within the Unreal Engine 5 (UE5) game engine. This process allowed for the visualisation of the recorded dance fragment in the virtual three-dimensional space.

Observing the prerecorded dance sequence evidenced the presence of the source body (the digital avatar, the performer) and the receptive body (the observer, the audience). This observation triggered the idea that the appreciation of this short dance sequence could be understood as if the dance were travelling from the source body towards the receptive body. The appreciation of a moving body could be metaphorically seen as if movement traces were emerging from a source body in the direction of a receptive body, which will receive the movement information as an echo of the dance travelling through space. This approach was taken and further explored using UE5.

After multiple experimentations and approaches to visualising movement traces were trialled, particle systems unveiled a way in which movement traces from a moving avatar could be directed towards a receptor body. A Niagara system, comprising particle emitters positioned on selected joints of the virtual dancer, facilitated the spawning (or creation) of particles at regular intervals. As the avatar moved, variations in the distance of the particles travelling towards the receptive body became apparent. These variations manifested as either wider spacing between particles (when the positioned emitter moved away from the target body) or particle clusters (when the positioned emitter moved closer to the target body) (Fig 1). This observation showed that particles created at constant and regular time intervals directed towards a target body inherently contained in their distribution a particular set of characteristics. These characteristics echo the position, velocity, acceleration, and overall manifestations of the moving body dynamics. Thus, different choreographies or movement sequences will generate a different and unique set of movement traces. In consequence, the particles that impact the receptive body will carry a unique configuration of rhythmical collisions and positions on the receptive body, depending on the movement characteristics.

Even though the aim is the development of non-visual experiences, the game engine works as a supportive intermediary space of exploration to work with movement data. It proved to be an exciting approach for experimentation in the transformation of movement characteristics into echoes that could be perceived as rhythmical collisions on a receptive body.

4.2 Haptic Scores to Haptic Experiences

Using a template of the position of the Teslasuit's electrodes and reproducing the approach from UE5 into Rhinoceros 3D, it is possible to map the collision area with the corresponding electrode. A reference image showing the distribution of haptic channels serves as a guide for positioning haptic effects in the different sections of the suit. In this way, it is viable to create a score of collisions mapped to Teslasuit (Fig. 2). This registration of movement traces on the target body can be used as a source for creating a haptic sequence in Teslasuit's Studio software.

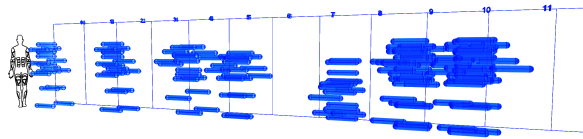


Figure 2: Rhinoceros 3D screenshot displaying TeslaSuit's haptic channels template (left), the geometries representing the particles/collision/haptic effect distributed on time.

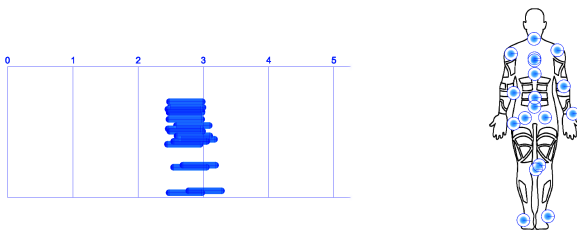


Figure 3: Rhinoceros 3D screenshot displaying TeslaSuit's haptic channels template (right), the geometries representing the particles/collision/haptic effect distributed on time (here 0-5 seconds). We can observe how the collisions are distributed on the TeslaSuit layout (right) and the start and duration of the collision in time (left).

Observing the sequence of collisions at a particular time, we can then transfer the information to the TeslaSuit's Studio software, adding the desired haptic effect to its corresponding place and duration (Fig. 3). After this, we will have a haptic score with the same duration as the original movement capture. The haptic scores created through the described methodology of data manipulation from Rhinoceros 3D and Grasshopper into TeslaSuit's Studio software can now be played with the embedded TeslaSuit's Studio haptic player (Fig. 4). Similar to a voice or camera recorder, the haptic player allows users to play, pause, or stop the haptic animation on the paired and connected TeslaSuit. This haptic animation offers the TeslaSuit's wearer a stand-alone and non-visual dependent haptic experience.

5 Evaluation

A set of iterations engaging participants aims to optimise the methods and methodologies for translating movement data into haptic materials. During the iterations, quantitative data from TeslaSuit and qualitative data will be collected in the form of experience sampling [14], unstructured interviews [6], body maps [5], and participant observation. Each new iteration will provide valuable information that will help improve the design of interactions and outcomes, and this will be evaluated through the researcher's reflective practice [22] and a thematic analysis of the participant's

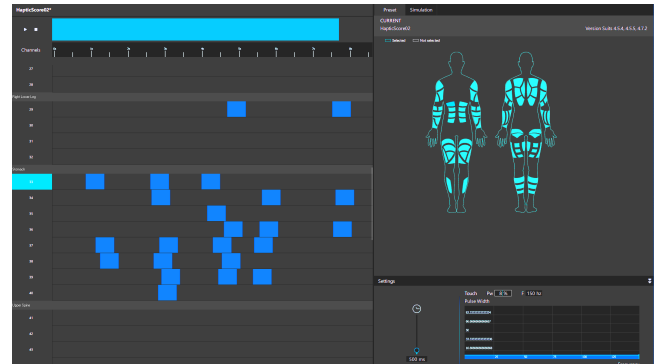


Figure 4: TeslaSuit's Studio software screenshot displaying the translation from the haptic scores into the distribution of haptic sensations to be triggered by TeslaSuit.

qualitative data. The thematic analysis is divided into three main topics, (a) meeting with technological devices and settings, (b) the interactive experience, (c) expectations and recommendations. The result of the analysis is then used to generate a new, updated experience that will generate a new iteration. This process will last until the beginning of the third year of the candidature (until the end of the placement at the host institution). After that, a period of reflection, documentation and design of a final reproducible method will follow at Coventry University, together with the writing of findings and dissertation.

6 Expected Contributions

This practice of working with human motion to create non-visual aesthetics has prompted different ways in which to understand choreographic processes, shifting the understanding of how to create movement that is not to be seen but to be sensed. Simultaneously, this methodology generates the possibility of using haptic sensation as an experience to influence a dancer's movement. A dancer wearing a TeslaSuit can respond to the haptic score of their own previous dance or that of someone else who is not physically present. This approach then expands towards new forms of creativity that can influence current and future artistic practices. The innovative aspect of this research lies in the fact that the purpose of this dance scoring is not to be reenacted but to be experienced, not as a memory of a performance but as an interaction. This approach could potentially enable visually impaired individuals to perceive other artistic manifestations. It could also inspire new ways of producing somatic experiences in physical spaces while interacting with digital ones, transcending reliance on screens or visual input, and benefiting a range of fields from artistic practices to game design. The collected quantitative and qualitative data from participants, along with its analysis, are expected to serve as a valuable source to inform future research on the development and use of electrical stimulation. This research will offer a critical perspective on the use and ethical considerations of EMS and TENS in artistic practices. Above all, it shifts the focus from attempting to accurately replicate familiar sensations to using haptic technology as a gateway for discovering, exploring, and experiencing physically previously unimaginable realms.

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